

Exercise 4: Arrange-Act-Assert (AAA) Pattern, Test Fixtures, Setup and Teardown Methods in JUnit

BankAccount.java

```
package com.example;

public class BankAccount {

    private double balance;

    public BankAccount(double initialBalance) {
        this.balance = initialBalance;
    }

    public double getBalance() {
        return balance;
    }

    public void deposit(double amount) {
        balance += amount;
    }
}
```

BankAccountTest.java

```
package com.example;

import org.junit.jupiter.api.AfterEach;
import org.junit.jupiter.api.BeforeEach;
import org.junit.jupiter.api.Test;

import static org.junit.jupiter.api.Assertions.assertEquals;

public class BankAccountTest {

    private BankAccount account;

    @BeforeEach
    void setUp() {
        account = new BankAccount(100);
        System.out.println("Before Test: Account initialized with $100.");
    }
}
```

```

@AfterEach
void tearDown() {
    account = null;
    System.out.println("After Test: Cleaning up account object.");
}

```

```

@Test
void testDeposit() {

    // Arrange
    double amount = 50;

    // Act
    account.deposit(amount);

    // Assert
    assertEquals(150, account.getBalance());
    System.out.println("Test: Deposit logic verified.");
}
}

```

Output:

